

Machine Learning Fundamentals

Sample educational document created for testing IntelliAssist AI.

1. What is Machine Learning?

Machine Learning (ML) is a branch of AI in which algorithms learn patterns from data and use those patterns to make predictions or decisions. Instead of specifying every rule manually, developers provide examples and an objective that guides learning. ML is commonly used for classification, regression, clustering, recommendation, anomaly detection, and ranking.

2. Supervised Learning

In supervised learning, the training dataset contains examples with known target values. A classification model predicts a category such as spam or not spam, while a regression model predicts a numerical value such as demand or price. Common supervised algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, and neural networks. Model quality should be measured using metrics appropriate to the task.

3. Unsupervised Learning

Unsupervised learning works with data where explicit target labels are not provided. Clustering methods group similar observations, while dimensionality-reduction techniques create compact representations of high-dimensional data. These methods can help discover structure, customer segments, topics, or unusual patterns. Because there may be no single correct answer, interpretation is an important part of evaluation.

4. Reinforcement Learning

Reinforcement learning describes a setting in which an agent interacts with an environment and receives rewards or penalties for actions. The goal is to learn a policy that maximizes cumulative reward. Reinforcement learning is useful for sequential decision problems such as game playing, robotics, resource allocation, and control. Exploration and exploitation are central challenges: the agent must balance trying new actions with using actions that already appear effective.

5. Training, Validation, and Testing

A reliable ML workflow separates data into training, validation, and test portions. Training data is used to learn model parameters. Validation data can guide model selection and hyperparameter tuning. Test data should be held back for final evaluation. Data leakage occurs when information from evaluation data improperly influences training, producing overly optimistic results.

6. Overfitting and Generalization

Overfitting occurs when a model learns details or noise specific to the training data and performs poorly on unseen examples. Underfitting occurs when the model is too simple to capture important patterns. Regularization, appropriate model complexity, more representative data, and careful validation can improve generalization. The goal is not to memorize the training set but to learn patterns that transfer to new inputs.

7. Machine Learning in Document AI

Document AI can use machine learning for classification, language understanding, similarity estimation, and information extraction. Embedding models convert text into numerical vectors so that semantically similar passages can be located efficiently. These representations are particularly useful for semantic search and retrieval-augmented generation systems.